

Trust Degradation Module (TDM)

OriginID: OOF-OID-ASGA-ARTS-TDM-2026-07-06-0003
Architecture Ecosystem: Structured Reality Standards™
Architecture Family: Autonomous Systems Governance Architecture (ASGA™)
Operational Layer: Autonomous Operational Governance Layer
Governed Space: Trust Degradation
Category: Governance & Enforcement
Subcategory: Autonomous Operational Governance
Type: Autonomous Risk & Trust Standard Module
Parent Standard: Autonomous Risk & Trust Standard (ARTS)
Version: 1.0
Status: Canonical · Open Module
Origin Date: 6 July 2026
Compatibility: OOF Methodology OS · GOA™ · ORA™ · AGA™ · AIG® · CLIA® · MGIA™
AI-Readable: Yes
Authority: OOF®
Protection: MIP™ — Methodological Intellectual Property
Canonical Language: English (UCL)

Canonical Definition

Trust Degradation Module (TDM) defines the structural conditions under which reductions in operational trust are identified, evaluated, monitored, and governed throughout autonomous operation.

TDM governs trust degradation.

The module establishes the governance conditions required to ensure that declining operational trust is detected early and addressed before it compromises autonomous operation.

Trust is not permanent.

Trust may increase or decrease over time.

TDM governs that degradation.

Module Operational Role

TDM defines the trust degradation layer of ARTS.

Its role is to detect, evaluate, and govern declining operational trust before it creates unacceptable operational risk.

Module Operational Space

TDM governs:

- trust degradation,
- declining operational trust,
- trust deterioration,
- trust monitoring,
- operational confidence reduction,
- governance trust degradation,
- trust status assessment,
- continuous trust observation.

Module Function

The module applies wherever organizations must detect and govern reductions in operational trust throughout autonomous execution.

Minimum Implementation Framework

1. Define the Trust Degradation Object

Identify which autonomous operational activities require monitoring for declining trust.

2. Define Trust Degradation Conditions

Establish governance conditions for identifying and evaluating reductions in operational trust.

3. Define Trust Degradation Failure Logic

Identify conditions where declining trust is not detected, incorrectly evaluated, or governance-invalid.

4. Define Governance Response

Define governance actions for increased monitoring, operational restriction, trust review, escalation, or corrective action.

5. Preserve Trust Traceability

Preserve trust assessments, governance decisions, operational evidence, degradation history, and supporting documentation.

Use Case 1 — Autonomous Inspection Drone

Scenario

An autonomous inspection drone begins producing inconsistent inspection results during routine operations.

Application

TDM detects the reduction in operational trust and initiates additional governance review.

Result

Potential operational issues are identified before trust deteriorates to unacceptable levels.

Use Case 2 — Enterprise Autonomous AI Agent

Scenario

An autonomous AI agent gradually produces recommendations with decreasing reliability.

Application

TDM monitors trust degradation and triggers governance actions before operational confidence is lost.

Result

The organization preserves trustworthy autonomous operation through early governance intervention.

Canonical Closing Statement

Trustworthy autonomy requires continuous awareness of declining trust. Trust Degradation ensures that reductions in operational confidence are identified, governed, and addressed before they undermine safe and reliable autonomous operation.

Related Documents

→ [Autonomous Risk & Trust Standard \(ARTS\)](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>